

ZAMOKTA, Russian Federation

WMO: 307410

Lat: 52.767N

Lon: 109.967E

Elev: 4219

StdP: 12.59

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.2	(c) -20.7	(d) -30.3	(e) 1.2	(f) -22.4	(g) -26.4	(h) 1.5	(i) -19.9	(j) 29.5	(k) -10.7	(l) 24.8	(m) -12.3	(n) 8.5	(o) 300	(p) 0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.6	(c) 77.8	(d) 58.8	(e) 74.6	(f) 57.4	(g) 71.5	(h) 56.8	(i) 62.7	(j) 71.2	(k) 60.7	(l) 69.2	(m) 58.9	(n) 67.0	(o) 6.8	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.9	90.2	66.2	57.9	83.7	63.7	55.8	77.5	62.0	30.5	71.1	28.9	69.3	27.6	66.8	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	19.3	16.9	-28.1	83.8	6.0	3.7	-32.4	86.5	-35.9	88.6	-39.3	90.7	-43.6	93.4
			-27.7	65.8	5.9	2.1	-31.9	67.3	-35.4	68.6	-38.7	69.8	-43.1	71.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	-5.0	0.4	13.4	28.3	41.1	55.5	60.4	55.8	42.5	26.5	9.7	-0.4		
	DBStd	24.34	8.67	10.39	11.00	9.24	8.82	8.10	5.92	6.64	8.60	9.73	9.69	8.27		
	HDD50	9006	1706	1390	1134	653	302	38	3	23	258	727	1210	1562		
	HDD65	13730	2171	1810	1599	1102	742	296	166	290	675	1192	1660	2027		
	CDD50	790	0	0	0	1	25	203	325	203	33	0	0	0		
	CDD65	41	0	0	0	0	1	11	23	6	0	0	0	0		
	CDH74	322	0	0	0	0	11	121	160	29	1	0	0	0		
	CDH80	42	0	0	0	0	1	19	21	1	0	0	0	0		
	WSAvg	6.7	6.9	6.8	7.1	7.8	8.2	6.7	5.4	5.5	6.5	6.5	6.5	6.4		
(15) Precipitation	PrecAvg	14.2	0.3	0.2	0.2	0.6	0.9	1.9	3.5	2.9	1.7	0.7	0.7	0.6		
	PrecMax	19.5	0.8	0.4	0.7	1.3	2.3	5.2	8.4	5.9	3.6	2.6	1.1	1.3		
	PrecMin	8.7	0.1	0.1	0.1	0.1	0.3	0.4	0.7	1.0	0.4	0.2	0.3	0.3		
	PrecStd	3.1	0.2	0.1	0.1	0.3	0.5	1.2	1.7	1.3	0.8	0.5	0.2	0.2		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	28.4	43.3	58.6	74.6	82.7	83.1	77.8	70.9	54.3	36.3	20.4		
		MCWB	15.5	23.0	33.5	41.3	51.5	59.3	61.7	58.4	53.9	40.8	30.7	18.9		
	2%	DB	12.7	22.2	37.0	51.6	66.8	77.1	78.5	73.7	65.3	48.2	30.7	16.0		
		MCWB	10.9	18.2	29.0	38.1	48.2	57.7	60.2	57.6	51.2	37.6	27.0	14.6		
	5%	DB	9.5	18.4	32.3	47.0	61.7	73.9	75.3	70.5	60.2	43.6	27.2	13.1		
		MCWB	8.1	15.7	26.2	35.6	45.5	56.5	58.7	56.7	49.4	35.0	24.6	12.1		
	10%	DB	6.5	15.0	28.8	42.5	56.9	70.2	71.9	67.3	55.8	39.3	23.3	10.6		
		MCWB	5.3	12.8	23.6	33.2	42.8	55.0	58.1	56.1	47.1	32.6	21.6	9.7		
	0.4%	WB	15.7	23.1	34.5	42.4	53.2	66.9	65.7	64.7	56.9	42.5	30.8	19.1		
		MCDB	17.3	28.4	42.1	56.9	72.5	74.6	75.9	70.5	66.5	52.9	34.9	19.9		
	2%	WB	11.1	18.8	29.8	38.7	49.1	61.9	62.9	61.1	52.9	38.3	27.4	14.7		
		MCDB	12.5	21.8	36.5	50.1	64.9	71.2	72.1	67.7	62.3	46.7	30.2	15.6		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	8.3	15.8	26.7	36.3	46.2	58.9	61.2	58.7	50.5	35.5	24.5	12.2		
		MCDB	9.5	18.3	32.2	45.9	60.0	68.6	69.9	65.9	58.0	42.4	27.1	13.0		
	10%	WB	5.4	12.8	23.7	33.6	43.4	56.4	59.7	57.0	48.3	33.1	21.6	9.7		
		MCDB	6.5	15.0	28.6	42.0	55.6	66.9	68.0	65.0	54.8	38.7	23.5	10.5		
		MDBR	11.7	14.1	16.6	16.7	18.4	19.4	17.6	16.1	15.8	13.8	11.8	10.8		
		MCDBR	14.5	16.7	18.6	20.9	26.1	24.6	22.5	21.1	21.5	19.1	15.9	11.5		
(36) Mean Daily Temperature Range	5% DB	MCWBR	13.1	13.7	13.1	12.3	13.3	11.5	9.6	9.6	11.2	11.2	12.9	10.3		
		MCDBR	14.4	16.2	18.2	19.9	23.8	21.0	18.1	17.4	19.4	17.8	15.6	10.9		
	5% WB	MCWBR	13.1	13.4	13.4	12.3	13.3	13.1	10.0	10.2	11.5	11.3	13.2	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.255	0.283	0.329	0.396	0.407	0.377	0.343	0.307	0.276	0.252	0.231		
	taud		2.219	2.205	2.215	2.268	2.158	2.230	2.373	2.451	2.506	2.497	2.308	2.177		
	Ebn at Noon		247	274	288	287	270	268	275	280	280	268	236	221		
	Edh at Noon		20	28	35	38	44	42	36	31	26	21	18	17		
(44) All-Sky Solar Radiation	RadAvg		373	699	1166	1595	1811	1899	1750	1480	1120	694	381	268		
	RadStd		22	29	36	90	104	122	145	79	88	61	28	17		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	-3.11	-3.73	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page